

23.3 Bootstrap confidence intervals

A bootstrap distribution gives a confidence interval through its quantiles, in several ways. They agree to first order, which is all the consistency theorems guarantee, and differ in how fast their coverage approaches the nominal level.

Let se be a standard error for $\hat{\theta} = \mathbf{c}^\top \hat{\boldsymbol{\beta}}$, classical or sandwich, and let $\hat{\theta}^*$ and se^* be computed from a resample in the same way. Write $\hat{q}(\gamma)$ and $\hat{t}(\gamma)$ for the γ quantiles of the bootstrap distributions of $\hat{\theta}^*$ and of the **bootstrap t statistic** $T^* = (\hat{\theta}^* - \hat{\theta})/\text{se}^*$.

Definition 23.3.1. Percentile, basic and studentized intervals

The nominal $1 - \alpha$ bootstrap intervals for θ are:

- (a) the **percentile** interval $[\hat{q}(\alpha/2), \hat{q}(1 - \alpha/2)]$;
- (b) the **basic** interval $[2\hat{\theta} - \hat{q}(1 - \alpha/2), 2\hat{\theta} - \hat{q}(\alpha/2)]$;
- (c) the **studentized** (or **bootstrap- t**) interval $[\hat{\theta} - \hat{t}(1 - \alpha/2)\text{se}, \hat{\theta} - \hat{t}(\alpha/2)\text{se}]$.

The basic interval translates the bootstrap idea directly: if $\hat{\theta} - \theta$ had the distribution of $\hat{\theta}^* - \hat{\theta}$, then $\hat{q}(\alpha/2) - \hat{\theta} \leq \hat{\theta} - \theta \leq \hat{q}(1 - \alpha/2) - \hat{\theta}$ would have probability $1 - \alpha$, and solving for θ gives (b). The studentized interval does the same with the root $(\hat{\theta} - \theta)/\text{se}$, as the t interval of Corollary 7.5.2 does. The percentile interval looks naive, but if some increasing ϕ makes $\phi(\hat{\theta}) - \phi(\theta)$ a pivot symmetric about zero, it is exact without knowing ϕ ([Exercise \(B1\)](#)). It is also **transformation respecting**, because quantiles commute with increasing functions; the basic and studentized intervals are not.

23.3.1 First-order correctness

The consistency theorems translate into coverage through a lemma about quantiles.

Lemma 23.3.1. Consistent bootstrap quantiles

Let R_n be random variables with distribution functions H_n , and let $\hat{H}_n$ be random distribution functions with $\sup_x |\hat{H}_n(x) - H_n(x)| \rightarrow 0$ in probability. Suppose $H_n(x) \rightarrow H(x)$ for every x , where H is continuous and strictly increasing on $\mathbb{R}$. For $0 < \gamma < 1$, let $\hat{\xi}_n(\gamma) = \inf\{x : \hat{H}_n(x) \geq \gamma\}$ and $\xi(\gamma) = H^{-1}(\gamma)$. Then $\hat{\xi}_n(\gamma) \rightarrow \xi(\gamma)$ in probability, and $\Pr(R_n \leq \hat{\xi}_n(\gamma)) \rightarrow \gamma$.

Proof. By Pólya's theorem $\sup_x |H_n(x) - H(x)| \rightarrow 0$, so $\delta_n = \sup_x |\hat{H}_n(x) - H(x)| \rightarrow 0$ in probability. Fix $\epsilon > 0$, write $\xi = \xi(\gamma)$, and let $\kappa = \min\{\gamma - H(\xi - \epsilon), H(\xi + \epsilon) - \gamma\} > 0$. On the event $\delta_n < \kappa$, $\hat{H}_n(\xi - \epsilon) < \gamma$ and $\hat{H}_n(\xi + \epsilon) > \gamma$, so $\xi - \epsilon \leq \hat{\xi}_n(\gamma) \leq \xi + \epsilon$. This proves the first claim. For the second,

$$\begin{aligned} \Pr(R_n \leq \hat{\xi}_n) &\geq H_n(\xi - \epsilon) - \Pr(|\hat{\xi}_n - \xi| > \epsilon), \\ \Pr(R_n \leq \hat{\xi}_n) &\leq H_n(\xi + \epsilon) + \Pr(|\hat{\xi}_n - \xi| > \epsilon). \end{aligned}$$

Let $n \rightarrow \infty$ and then $\epsilon \rightarrow 0$, using the continuity of H at ξ . ■

Corollary 23.3.2. Bootstrap intervals are first-order correct

Under the conditions of Theorem 23.1.4, the percentile and basic intervals built from the residual bootstrap have coverage probability tending to $1 - \alpha$.

Proof. Take $R_n = (\hat{\theta} - \theta)/\|\mathbf{a}_n\|$, with $\hat{H}_n$ the bootstrap distribution of $(\hat{\theta}^* - \hat{\theta})/\|\mathbf{a}_n\|$ and $H(x) = \Phi(x/\sigma)$. Theorem 23.1.4 gives the hypotheses of Lemma 23.3.1. The basic interval covers θ if and only if $\hat{q}(\alpha/2) - \hat{\theta} \leq \hat{\theta} - \theta \leq \hat{q}(1 - \alpha/2) - \hat{\theta}$, that is, $\hat{\xi}_n(\alpha/2) \leq R_n \leq \hat{\xi}_n(1 - \alpha/2)$ with $\hat{\xi}_n$ the quantiles of $\hat{H}_n$. By the lemma the probability tends to $(1 - \alpha/2) - \alpha/2$, the lower endpoint contributing no mass in the limit because H is continuous. The percentile interval covers θ if and only if $-\hat{\xi}_n(1 - \alpha/2) \leq R_n \leq -\hat{\xi}_n(\alpha/2)$. Since H is symmetric, $-\xi(1 - \alpha/2) = \xi(\alpha/2)$, and the same argument applies. ■

The same argument applies to the wild bootstrap under the conditions of Theorem 23.2.2. Since τ_n need not converge, take $R_n = (\hat{\theta} - \theta)/(\|\mathbf{a}_n\|\tau_n)$ and $H = \Phi$; dividing by the constant τ_n leaves the intervals unchanged. It applies also to studentized intervals whenever the standard error is consistent. None of this separates the methods from one another or from the normal interval $\hat{\theta} \pm z_{\alpha/2} \text{se}$.

23.3.2 Pivots and higher-order accuracy

The case for studentizing is clearest when the root is an exact pivot.

Proposition 23.3.3. The parametric bootstrap in the normal model

In the normal linear model with $\text{rank}(\mathbf{X}) = p < n$, let the **parametric bootstrap** draw $\mathbf{Y}^* \sim N_n(\mathbf{X}\hat{\boldsymbol{\beta}}, s^2\mathbf{I})$. Then for every data set:

- (a) $T^* = (\mathbf{c}^\top \hat{\boldsymbol{\beta}}^* - \mathbf{c}^\top \hat{\boldsymbol{\beta}})/(s^* \|\mathbf{a}\|) \sim t(n - p)$ under Pr_* , so the studentized interval is the t interval of Corollary 7.5.2, apart from Monte Carlo error;
- (b) $\mathbf{c}^\top \hat{\boldsymbol{\beta}}^* \sim N(\mathbf{c}^\top \hat{\boldsymbol{\beta}}, s^2 \|\mathbf{a}\|^2)$ under Pr_* , so the percentile and basic intervals are $\hat{\theta} \pm z_{\alpha/2} s \|\mathbf{a}\|$, whose coverage is $\Pr(|T| \leq z_{\alpha/2}) < 1 - \alpha$ with $T \sim t(n - p)$.

Proof. Given the data, $\mathbf{Y}^*$ follows the normal linear model with coefficient vector $\hat{\boldsymbol{\beta}}$ and variance s^2 . (a) is Corollary 7.5.2 applied to that model, with true value $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$. (b) is Theorem 7.5.1(a) for the same model. The interval $\hat{\theta} \pm z_{\alpha/2} s \|\mathbf{a}\|$ covers θ if and only if $|T| \leq z_{\alpha/2}$, where $T = (\hat{\theta} - \theta)/(s \|\mathbf{a}\|) \sim t(n - p)$ by Corollary 7.5.2, and $t(n - p)$ has heavier tails than $N(0, 1)$. ■

The studentized root has the same distribution in both worlds, so the bootstrap reproduces it exactly. The unstudentized root depends on σ , and the bootstrap substitutes s , losing the spread due to estimating σ . The nonparametric bootstrap is never exact, but the same mechanism operates: an **asymptotic pivot** is reproduced more accurately than a root whose distribution

depends strongly on the unknown law. Hall (1988) made this precise with Edgeworth expansions. Under smoothness conditions the one-sided studentized interval has coverage error of order n^{-1} , the one-sided percentile and basic intervals of order $n^{-1/2}$. We do not prove these rates (see Hall 1992). The price of studentizing is a standard error recomputed in each resample, and instability when it is poorly estimated.

BCa. Efron (1987) corrected the percentile interval to the same order of accuracy without a standard error. The **bias-corrected and accelerated** (BCa) interval uses the percentile interval at adjusted levels

$$\gamma_k = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_k}{1 - \hat{a}(\hat{z}_0 + z_k)}\right), \quad z_1 = z_{(\alpha/2)}, \quad z_2 = z_{(1-\alpha/2)},$$

where $z_{(\gamma)} = \Phi^{-1}(\gamma)$, the **bias correction** $\hat{z}_0 = \Phi^{-1}(\Pr_*(\hat{\theta}^* < \hat{\theta}))$ measures median bias, and the **acceleration** $\hat{a}$, the rate at which the standard error changes with θ on a normalizing scale, is usually estimated from the jackknife values $\hat{\theta}_{(i)}$, computed without case i :

$$\hat{a} = \frac{\sum_i d_i^3}{6(\sum_i d_i^2)^{3/2}}, \quad d_i = \bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)}.$$

In regression no refitting is needed: $\hat{\theta}_{(i)} = \hat{\theta} - a_i \hat{\varepsilon}_i / (1 - h_{ii})$ by Proposition 10.6.2(a) ([Exercise \(B2\)](#)). With $\hat{z}_0 = \hat{a} = 0$ BCa is the percentile interval. Its derivation and second-order accuracy are in Efron (1987) and DiCiccio and Efron (1996), for resampling cases, to which the jackknife formula above belongs. Pairing it with the residual bootstrap, as below, is a common heuristic that Efron's proof does not cover.

23.3.3 An example and a simulation

Example. Intervals for the poverty coefficient

Return to the state murder rates of Example (Poverty and murder rates), with the District of Columbia omitted, and the coefficient 0.2538 of poverty in the regression on poverty, single parenthood and urbanization. Seven 95% intervals from $B = 9999$ resamples:

method	interval	length
t , classical standard error	(0.066, 0.442)	0.376
residual bootstrap, percentile	(0.064, 0.437)	0.374
residual bootstrap, basic	(0.070, 0.444)	0.374
residual bootstrap, studentized	(0.071, 0.449)	0.378
residual bootstrap, BCa	(0.040, 0.418)	0.378
case bootstrap, percentile	(0.080, 0.428)	0.348
wild bootstrap- t , HC2	(0.079, 0.428)	0.348

The residual bootstrap used leverage-adjusted residuals. The first four intervals agree closely: with fifty states and a well-behaved fit, the classical interval is hard to improve on. BCa is shifted down; its bias correction $\hat{z}_0 = -0.016$ is negligible, but the jackknife acceleration $\hat{a} = -0.0458$ moves both endpoints. The shift comes from that heuristic pairing, not from BCa as such, and opposes the small upward shift of the studentized interval. The case and wild bootstraps give shorter intervals, because the HC2 standard error 0.0869 is smaller than the classical 0.0934 for these data. All seven exclude zero.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")

rng = np.random.default_rng(2303)
B = 9999
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
XtX_inv = np.linalg.inv(X.T @ X)
A = XtX_inv @ X.T
a, c = A[1], np.sqrt(XtX_inv[1, 1]) # the poverty coefficient is a @ y
h = np.sum((X @ XtX_inv) * X, axis=1)
fit = X @ (A @ y)
e = y - fit
b = a @ y
s = np.sqrt(e @ e / (n - p))
q = [0.025, 0.975]
out = {"t": b + np.array([-1, 1]) * stats.t.ppf(0.975, n - p) * s * c}

r = e / np.sqrt(1 - h) # residual bootstrap, modified
residuals
r -= r.mean()
Ystar = fit + r[rng.integers(0, n, size=(B, n))]
bstar = Ystar @ a
sstar = np.sqrt(np.sum((Ystar - (Ystar @ A.T) @ X.T) ** 2, axis=1) / (n - p))
tstar = (bstar - b) / (sstar * c)
lo, hi = np.quantile(bstar, q)
out["percentile"] = np.array([lo, hi])
out["basic"] = np.array([2 * b - hi, 2 * b - lo])
out["studentized"] = b - np.quantile(tstar, q[::-1]) * s * c

z0 = stats.norm.ppf(np.mean(bstar < b)) # BCa
loo = b - a * e / (1 - h) # case-jackknife values (heuristic
with residual resampling)
dev = loo.mean() - loo
acc = np.sum(dev ** 3) / (6 * np.sum(dev ** 2) ** 1.5)
zs = z0 + stats.norm.ppf(q)
out["BCa"] = np.quantile(bstar, stats.norm.cdf(z0 + zs / (1 - acc * zs)))

idx = rng.integers(0, n, size=(B, n)) # case bootstrap
bcase = np.array([np.linalg.lstsq(X[i], y[i], rcond=None)[0][1] for i in idx])
out["case"] = np.quantile(bcase, q)

se2 = np.sqrt(np.sum(a ** 2 * e ** 2 / (1 - h))) # wild bootstrap-t with HC2
Ystar = fit + (e / np.sqrt(1 - h)) * rng.choice([-1.0, 1.0], size=(B, n))
Estar = Ystar - (Ystar @ A.T) @ X.T
```

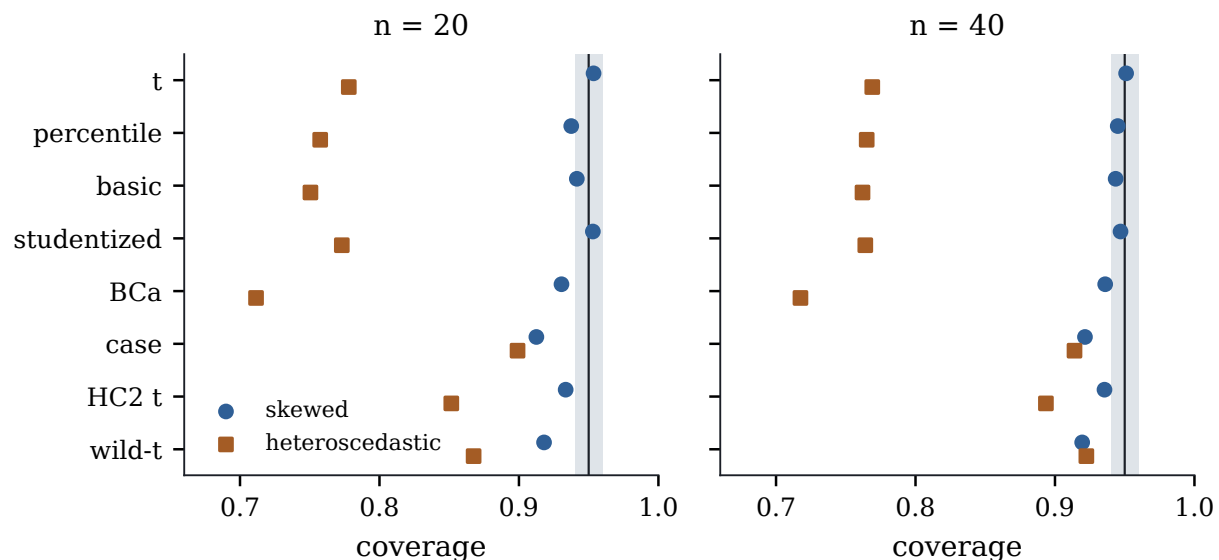


Figure 23.3.1: Coverage of eight nominal 95% intervals for a slope. Circles: skewed errors, constant variance. Squares: normal errors with variance increasing in x . The band is two Monte Carlo standard errors around 0.95.

```
wstar = (Ystar @ a - b) / np.sqrt((Estar ** 2 / (1 - h)) @ a ** 2)
out["wild-t"] = b - np.quantile(wstar, q[::-1]) * se2

for m, (l, u) in out.items():
    print(f"{m:12s} ({l:.3f}, {u:.3f})    length {u - l:.3f}")
```

[Run this code online](#)

One data set cannot show which interval is right. Figure 23.3.1 reports coverages for the slope of a line, from 2000 simulated data sets with $B = 999$ resamples each, and Monte Carlo standard error about 0.005. The fixed design is skewed, with largest leverage 0.433 at $n = 20$ and 0.309 at $n = 40$. The errors are either centred exponential with constant variance, or normal with standard deviation proportional to x , so that the largest variances sit at the high-leverage end.

Three lessons stand out.

1. *With constant variance, studentization is what matters.* At $n = 20$ the classical t interval (0.954) and the studentized residual bootstrap (0.953) are on target despite skewed errors. The percentile (0.938) and BCa (0.930) intervals are a little short, as Proposition 23.3.3 predicts for intervals that ignore the variability of s . The robust methods (case 0.912, wild- t 0.918) pay for estimating a variance that did not need estimating.
2. *With heteroscedastic errors the classical t interval and the four residual-bootstrap intervals all fail, and more data do not help:* all five cover between 0.71 and 0.78 at both sample sizes (Exercise (B1)).
3. *The robust methods improve with n but start below nominal.* The case bootstrap, the HC2 t interval and the wild bootstrap- t cover 0.899, 0.852 and 0.868 at $n = 20$, and 0.914, 0.893 and 0.922 at $n = 40$. A high-leverage point with a large error variance is hard for all of them, since one residual carries much of the information about the variance that matters (Example (The wild bootstrap for Engel's data)).

```
import numpy as np
from scipy import stats
```

```

level = 0.95
lo_q, hi_q = (1 - level) / 2, (1 + level) / 2
beta = np.array([1.0, 1.0])
methods = ["t", "percentile", "basic", "studentized", "BCa", "case", "HC2 t", "wild-
t"]

def make_design(n):
    """A skewed fixed design and the quantities every data set reuses."""
    x = np.exp(0.5 * stats.norm.ppf((np.arange(1, n + 1) - 0.5) / n))
    X = np.column_stack([np.ones(n), x])
    XtX_inv = np.linalg.inv(X.T @ X)
    A = XtX_inv @ X.T # beta_hat = A y
    h = np.sum((X @ XtX_inv) * X, axis=1) # leverages
    sd = x / np.sqrt(np.mean(x ** 2)) # heteroscedastic sd, mean square
    one
    return dict(n=n, x=x, X=X, A=A, h=h, a=A[1], c=np.sqrt(XtX_inv[1, 1]), sd=sd)

def errors(rng, law, d):
    if law == "skewed": # centred exponential, constant
        variance
        return rng.exponential(size=d["n"]) - 1.0
    return d["sd"] * rng.normal(size=d["n"]) # normal, sd proportional to x

def one_data_set(rng, law, d, B):
    """Return, for each method, whether its 95% interval covers the true slope."""
    n, x, X, A, h, a, c = (d[k] for k in ("n", "x", "X", "A", "h", "a", "c"))
    y = X @ beta + errors(rng, law, d)
    b = a @ y
    e = y - X @ (A @ y)
    s = np.sqrt(e @ e / (n - 2))
    cover = {"t": abs(b - beta[1]) <= stats.t.ppf(hi_q, n - 2) * s * c}
    # residual bootstrap from leverage-adjusted, centred residuals
    r = e / np.sqrt(1 - h)
    r -= r.mean()
    Ystar = X @ (A @ y) + r[rng.integers(0, n, size=(B, n))]
    bstar = Ystar @ a
    Estar = Ystar - (Ystar @ A.T) @ X.T
    tstar = (bstar - b) / (np.sqrt(np.sum(Estar ** 2, axis=1) / (n - 2)) * c)
    q_lo, q_hi = np.quantile(bstar, [lo_q, hi_q])
    t_lo, t_hi = np.quantile(tstar, [lo_q, hi_q])
    cover["percentile"] = q_lo <= beta[1] <= q_hi
    cover["basic"] = 2 * b - q_hi <= beta[1] <= 2 * b - q_lo
    cover["studentized"] = b - t_hi * s * c <= beta[1] <= b - t_lo * s * c
    # BCa: residual-bootstrap bias correction, case-jackknife acceleration (a
    heuristic pairing)
    z0 = stats.norm.ppf(np.clip(np.mean(bstar < b), 1 / B, 1 - 1 / B))
    loo = b - a * e / (1 - h) # slopes with one case deleted
    dev = loo.mean() - loo
    acc = np.sum(dev ** 3) / (6 * np.sum(dev ** 2) ** 1.5)
    zs = z0 + stats.norm.ppf([lo_q, hi_q])
    bca_lo, bca_hi = np.quantile(bstar, stats.norm.cdf(z0 + zs / (1 - acc * zs)))
    cover["BCa"] = bca_lo <= beta[1] <= bca_hi
    # case bootstrap percentile
    idx = rng.integers(0, n, size=(B, n))
    xs, ys = x[idx], y[idx]
    xc = xs - xs.mean(axis=1, keepdims=True)
    sxx = np.sum(xc ** 2, axis=1)
    ok = sxx > 1e-12 # drop the (rare) resamples with

```

```

one distinct x
bcase = np.sum(xc[ok] * ys[ok], axis=1) / sxx[ok]
cl, ch = np.quantile(bcase, [lo_q, hi_q])
cover["case"] = cl <= beta[1] <= ch
# HC2 t interval, and the wild bootstrap-t studentized by HC2
se2 = np.sqrt(np.sum(a ** 2 * e ** 2 / (1 - h)))
cover["HC2 t"] = abs(b - beta[1]) <= stats.t.ppf(hi_q, n - 2) * se2
v = rng.choice([-1.0, 1.0], size=(B, n))
Ystar = X @ (A @ y) + (e / np.sqrt(1 - h)) * v
Estar = Ystar - (Ystar @ A.T) @ X.T
wstar = (Ystar @ a - b) / np.sqrt((Estar ** 2 / (1 - h)) @ a ** 2)
w_lo, w_hi = np.quantile(wstar, [lo_q, hi_q])
cover["wild-t"] = b - w_hi * se2 <= beta[1] <= b - w_lo * se2
return cover

def coverage(law, n, reps, B, seed):
    rng = np.random.default_rng(seed)
    d = make_design(n)
    hits = {m: 0 for m in methods}
    for _ in range(reps):
        for m, ok in one_data_set(rng, law, d, B).items():
            hits[m] += ok
    return {m: hits[m] / reps for m in methods}

for law in ["skewed", "heteroscedastic"]:
    cov = coverage(law, n=20, reps=200, B=499, seed=2331)
    print(f"[law:16s]" + " ".join(f"{m} {v:.2f}" for m, v in cov.items()))

```

[Run this code online](#)

With 200 data sets the cell's coverages have Monte Carlo standard errors of about $\sqrt{0.95 \times 0.05 / 200} = 0.015$.

23.3.4 Bootstrap tests of a linear hypothesis

A test of $H : \Lambda^\top \beta = \mathbf{d}$ can take its critical value from the bootstrap instead of the F table. In each residual-bootstrap resample from the full model, compute F_H^* of Eq. 11.3.1 for the hypothesis $\Lambda^\top \beta = \Lambda^\top \hat{\beta}$, which is true in the bootstrap world, and report the proportion of resamples with $F_H^* \geq F_H$. This is the testing counterpart of the studentized interval (Proposition 12.1.2). Alternatively, resample the residuals of the restricted fit, so that the null hypothesis holds in the bootstrap world; that is the bootstrap analogue of the Freedman–Lane scheme of [Section 23.5](#).

23.3.5 Exercises

23.3.5.1 A. Check your understanding

Exercise. A1

Show that the percentile and basic intervals coincide when the bootstrap distribution of $\hat{\theta}^*$ is symmetric about $\hat{\theta}$. With $\hat{\theta} = 2$, $\hat{q}(0.025) = 1.5$ and $\hat{q}(0.975) = 3.5$, compute both intervals and explain why they lean in opposite directions.

Solution. Symmetry about $\hat{\theta}$ means $\hat{q}(\gamma) - \hat{\theta} = \hat{\theta} - \hat{q}(1 - \gamma)$, so $2\hat{\theta} - \hat{q}(1 - \alpha/2) = \hat{q}(\alpha/2)$, and similarly for the other endpoint. In the example the percentile interval is $[1.5, 3.5]$ and the basic interval is $[4 - 3.5, 4 - 1.5] = [0.5, 2.5]$. The bootstrap distribution has a long right tail. The percentile interval copies it. The basic interval reflects it, reasoning that if $\hat{\theta}$ tends to fall far above θ with some probability, then θ may lie far below $\hat{\theta}$.

23.3.5.2 B. Practice

Exercise. B1

Suppose there is a strictly increasing ϕ such that $\phi(\hat{\theta}) - \phi(\theta)$ has a continuous distribution K , symmetric about zero and the same for every θ , and suppose the bootstrap reproduces it exactly: $\phi(\hat{\theta}^*) - \phi(\hat{\theta})$ has distribution K under Pr_* . Show that the percentile interval has coverage exactly $1 - \alpha$.

Solution. Let k_γ be the γ quantile of K , so $k_{\alpha/2} = -k_{1-\alpha/2}$. Under Pr_* , $\phi(\hat{\theta}^*) = \phi(\hat{\theta}) +$ a K -variable, so the γ quantile of $\hat{\theta}^*$ is $\hat{q}(\gamma) = \phi^{-1}(\phi(\hat{\theta}) + k_\gamma)$. Hence $\hat{q}(\alpha/2) \leq \theta \leq \hat{q}(1 - \alpha/2)$ if and only if $k_{\alpha/2} \leq \phi(\theta) - \phi(\hat{\theta}) \leq k_{1-\alpha/2}$, that is, $-k_{1-\alpha/2} \leq \phi(\hat{\theta}) - \phi(\theta) \leq -k_{\alpha/2}$. By symmetry this is $k_{\alpha/2} \leq \phi(\hat{\theta}) - \phi(\theta) \leq k_{1-\alpha/2}$, which has probability $1 - \alpha$.

Exercise. B2

Show that the jackknife values used in the BCa acceleration satisfy $\hat{\theta}_{(i)} = \hat{\theta} - a_i \hat{\varepsilon}_i / (1 - h_{ii})$, where $\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}$. Deduce that the jackknife variance estimate $\frac{n-1}{n} \sum_i (\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)})^2$ is close to the HC3 variance $\sum_i a_i^2 \hat{\varepsilon}_i^2 / (1 - h_{ii})^2$. (Chapter 21 shows that the uncentred sum $\sum_i (\hat{\theta}_{(i)} - \hat{\theta})^2$ equals it exactly.)

23.3.5.3 C. Going deeper

Exercise. C1

The coverage of a bootstrap interval can itself be estimated by a second level of bootstrapping: treat each first-level resample as data, and ask how often its interval covers $\hat{\theta}$. Describe how this can be used to recalibrate the nominal level, and count the least squares fits needed with $B_1 = B_2 = 999$. How does the design-fixed residual bootstrap reduce the cost?